Imperial College London

Department of Aeronautics

Group Design Project
Information for Students and Supervisors: 2024-25

MEng Aeronautical Engineering (H401)
MEng Aeronautics with Spacecraft Engineering (H415)

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1. Introduction

The third year large Group Design Project is a significant part of the H401 & H415 programmes, worth 25% of the third year's marks and occupying five full-time weeks immediately following the end of the third year summer exams.

It provides students with the opportunity to learn how to function and contribute creatively in a large multi-disciplinary team, simulating the industrial approach to engineering design and teamwork. With five different challenging projects on offer, students must develop innovative solutions, expanding on their prior knowledge and researching new ideas, while appreciating the constraints imposed in a multi-disciplinary design environment.

2. Contacts

Administrative:

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Academic Coordinator:

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3. Role Selection and Allocation

3.1 The 2024-2025 Group Design Projects

Neptune Exploration Mission (NEM25)

Neptune, the furthest confirmed major planet in the Solar System, offers an invaluable opportunity for scientific discovery. Due to its vast distance, Neptune has only been visited once during a flyby by the Voyager 2 spacecraft. Neptune is a planet shrouded in mystery. Winds in its atmosphere are supersonic and reach speeds of more than 2,200 km/h. It is heated internally due to mechanisms that are not well understood, and its magnetic field is curiously tilted. Neptune's largest moon, Triton, is one of the most geologically active Solar System bodies and it might have a subsurface ocean (maybe harboring life?).

The aim of this project is to design the first robotic space exploration mission to explore the Neptunian system and the Kuiper belt. A key element of the mission is a probe that will be deployed from the main spacecraft to descend through Neptune's atmosphere and investigate Neptune's core to gather data. The project team must address all aspects of space mission design, including project management, interaction with key stakeholders, definition and exploration of mission concepts and architectures, and spacecraft and individual subsystem design. This will involve overcoming considerable challenges associated with reaching the furthest planet in the Solar System, making sure that the mission operates reliably, and staying within the budget (sometimes more an art than a science).

Academic Advisors: Drs Amato & Bruce (coordinators), Cantwell, Lubbock, and Profs Aliabadi, Doorly, Laizet, Lee and Montomoli.

Roles Available:

- NEM01 – Project coordination & budget (x4)
- NEM02 – Neptune descent probe subsystem design (x8)
- NEM03 – Neptune orbiter subsystem design (x8)
- NEM04 – Trajectory design (x5)
- NEM05 – Launch & Interplanetary cruise systems (x5)

H2-STOL (HST25)

By the end of the project your team will have designed a 9 passenger hydrogen fuel-cell-powered short take-off and landing aircraft. The aircraft will be designed to operate for transportation and tourist operations in Canada, Alaska, New Zealand and remote regions of South East Asia. It should have a take-off and landing distance of around 250 m and a range of 700 nmi.

Oh, and we would also like it to be able to operate from lakes and other still bodies of water.

Academic Advisors: Dr Li, Liem, Panesar and Professors Chernyshenko, Hewson (coordinator), Mainini, Morrison and Palacios

Roles Available:

- HST01: Configuration Definition (x3)
- HST02: Propulsion System (x3)
- HST03: Aerodynamic Design (x2)
- HST04: Structural Design and Manufacturing (x3)
- HST05: Flight Dynamics and Control (x3)
- HST06: Safety, Systems and Ground Infrastructure (x3)
- HST07: Hydrodynamics (x3)

Shell Eco Marathon (SEM25)

The aim of the GDP is to design a 3-wheel prototype ultra-efficient, lightweight battery electric vehicle to compete in Shell Eco-marathon Europe and Africa. The competition takes place in the Siselia ring track in Polland. The objective is to complete 11 laps of the track (15.620 km) at an average speed above 25 km/h, i.e. in a time within 39 minutes.

The design will be a tadpole configuration (2 front wheels for steering and 1 rear wheel for power), which will use lithium-ion batteries and upgrade an existing chassis. The main purpose of the upgrade is to improve the aerodynamic performance by reducing flow separation, and to incorporate a small ventilator for driver comfort and, potentially, flow control. The final design should cost £20—25K (Imperial Racing Green Team allocated SEM budget).

The rules of the competition are available at: https://www.shellecomarathon.com/about/global-rules.html

Academic Advisors: Drs Billoti, MacCarthy, Tagarielli, Venetsanos, and Profs Papadakis, Peiró (coordinator).

Roles Available:

- SEM01 – Technical director: project coordination, race strategy and rules compliance (x1)
- SEM02 – Packaging, Integration and CAD (x3)
- SEM03 – Aeroshell aerodynamics (x4)
- SEM04 – Aeroshell structural analysis and manufacturing (x4)
- SEM05 – Powertrain (x2)
- SEM06 – Lap simulation, vehicle dynamics and traction (x2)
- SEM07 – Dashboard, communications, and telemetry (x1)
- SEM08 – Steering and braking (x1)
- SEM09 – Driver environment, ergonomics, and interfaces (x2)

HELICOPTER EMERGENCY RESCUE (HER25)

Helicopter Emergency Medical Services (HEMS) are crucial for providing rapid medical response, especially in situations where ground transport is too slow or inaccessible. HEMS plays a vital role in saving lives by quickly reaching accident scenes, transporting critically ill patients, and ensuring timely medical interventions. It is particularly essential in rural, mountainous, or congested urban areas where rapid evacuation can make the difference between life and death.

Current HEMS aircraft are existing helicopters retrofitted for medical missions. While highly capable, they have design drawbacks that limit their effectiveness, such as limited cabin space, noise and vibration issues, performance limitations (in hot-and-high conditions or adverse weather), range constraints, operating and maintenance costs, safety concerns and environmental constraints (e.g. noise levels).

The aim of this project is to design the perfect rotorcraft for HEMS operations, providing optimised cabin layout for medical operations, enhance safety features, performance improvements, noise and vibration reduction, cost and maintenance efficiency and all-weather operations.

Academic Advisors: Drs Armanini, Fasel, Ribera-Vicent (coordinator), Seifikar, Shamsuddin and Steiros.

Roles Available:

- HER01 – Project coordination (x2)
- HER02 – Aerodynamics (x5)
- HER03 – Structural Design, Manufacturing and CAD (x4)
- HER04 – Systems, Propulsion and Mission equipment(x5)
- HER05 – Flight Mechanics, Performance & Control (x4)

NEXT GENERATION TRAINER (NGT25)

The aim of this project is the development of a next-generation intermediate/advanced trainer aircraft for the Royal Air Force which can deliver the performance necessary to train aviators to fly next-gen fighters, at significantly lower emission and operating costs.

The resulting aircraft must meet all relevant airworthiness and safety requirements. It must accommodate two flight crew, achieve at least transonic speeds and be highly maneuverable. Provisions for an armed, multirole light attack variant, should be made.

Academic Advisors: Drs Eastham, Levis (coordinator), Quino Quispe and Rigas, Profs Robinson and Vincent

Roles Available:

- NGT01 – Project coordination (x2)
- NGT02 – Aerodynamics (x4)
- NGT03 – Structural Design, Manufacturing & CAD (x4)
- NGT04 – Systems & Propulsion (x4)
- NGT05 – Flight Mechanics, Performance & Control (x4)
- NGT06 – Flight Simulation & Testing (x2)

3.2 How to Make Your Selections – The Rules

Student selections will be made via the dedicated Sharepoint form no later than **Monday 28th April 2025**.

https://imperiallondon.sharepoint.com/sites/foe/aero/student-portal/Lists/GDP%20Role%20selection/AllItems.aspx

Indicate your choices by entering the *codes* for the roles you are interested in, in order of preference. Please make sure that you make your choice of roles carefully as mistakes cannot be rectified after the allocation has been made.

Students on the H401 Aeronautical Engineering programme may submit up to 20 choices. To ensure that all students receive one of their chosen projects, you must select at least two roles from each of the five projects.

Students on the H415 Spacecraft Engineering programme may only choose from roles in the space project (NEM0X) indicating what their order of preference of all 5 available roles is.

The allocation will be broadly based on students' past performance in the course, while trying to maximise student satisfaction and making sure all roles are filled. The choices of students who fail to follow the rules above will be ignored at first pass, and will be allocated whatever role remains.

4. During the Project

The projects will start on Monday 12th May 2025. Each team will hold a kick-off meeting, to be scheduled in discussion with the supervising academics, where students will be given more information about the project scope and aims. Students should also take this opportunity to meet with their individual supervisors to clarify their individual objectives within the project and set goals for the following week.

Each group will be assigned a room in which they can work and hold meetings. Should there be a shortage of available rooms, groups may have to share a single room and should therefore coordinate their activity timetables appropriately.

4.1 Team Organization

Although smaller sub-groups or individual students may be responsible for only one disciplinary aspect of the design, the entire group's success in this multidisciplinary design project relies on effective communication and collaboration of all group members.

In recognition of this, each group will typically have some students assigned or elected to fill a project management role. These students will act as coordinators to facilitate the running of the project. They will also act as mediators to solve design conflicts that may arise between individual member's design choices, but if an external decision is needed, it will need to be taken by the whole team through an appropriate voting mechanism.

To ensure that all students are kept aware of the group's progress, and how that might have an impact on their areas of responsibility, it is recommended that a short update meeting is held daily, where progress can be presented, design decisions discussed and new goals can be set.

4.2 Weekly Progress Meetings

Each team is expected to hold a weekly progress meeting where all group members and supervising academics will be in attendance. These meetings will be called by the project managers, who will organize the agenda according to the issues that need discussion. A schedule should be available to the academic advisors at the start. Times and dates are to be set in discussion with the supervising academics.

These meetings will offer both students and academics the opportunity to review the past week's progress and discuss any significant problems with their colleagues and staff, in order to reach generally acceptable decisions.

Students will prepare presentations for the weekly group meeting and will be given 10 minutes to present, followed by a 5-minute Q&A. Note that, not all students need to present each week, however each group member must have presented at least twice by the end of the project.

The content of these presentations will in part inform each student's Carrying Out mark, but will also offer an opportunity for students to receive feedback on their work and presentation skills. In addition to the feedback received during these presentations, students should aim to meet with their individual supervisors once a week, following the progress meetings or at a mutually convenient time.

4.3 Budget/Purchasing Guidelines

Each group has been given a £350 budget for any materials, printing or other resources they require. Please note that:

- The budget is meant for the purchase of consumables or other resources required for the successful completion of your project. It may not be used for food and drink.
- Each group should have no more than two people responsible for requesting the Purchase Orders or buying and submitting Expense Claim forms.
- The use of personal Credit Card may be allowed for purchase up to £100. Written approval (via email) from your supervisor is required before purchasing.
- All Expense Claim MUST BE submitted within 3 months of cost incurred.

- Before submitting an expense claim ensure all receipts are itemised; invoice or receipts are attached (credit card slips or bank statement alone will not be sufficient).
- If the group would like to buy items for the amount above £ 100 with suppliers registered on the College system, please send the Purchase Order request email to aero.finance@imperial.ac.uk.
- In exceptional situation, if the Group (person) need to buy anything for the amount above £100 and the supplier is not registered with the College the request for Virtual Credit Card can be sent to aero.finance@imperial.ac.uk. Please check the guidance.
- Account code – quote Project and Group (if applicable).

In addition to the above budget for consumables, each team will be allocated a budget of £15 per person to be used towards a single team building meal.

The linked documents will be helpful when requesting a virtual credit card, requesting a purchase order to be raised, and filling expense claims.

If you have any questions regarding purchasing items for your project, please contact **aero.finance@imperial.ac.uk**

5. Assessment

5.1 Project Outcomes / Deliverables

Students are expected to deliver the following pieces of work by the end of the project to report the project outcome and their contributions towards it.

- **Design specification document:** A single collaborative document which details the overall results of the project and the contributions of each subgroup towards achieving them. It can be considered as a summary of the entire project to which each member contributes and should be no longer than 5 pages in total.
- **Poster:** An A1 landscape poster or equivalent video, presenting a top level view of the projects result, detailing in an approachable way the most important design features and characteristics.
- **Individual report:** A technical report by each member of the group detailing their contribution to the project.
- **Group presentation:** A group technical presentation, presenting, detailing and discussing the resulting design.

Some projects may require additional deliverables such as a live demonstration, or demonstration videos from particular sub-groups.

5.2 Overall Project Weighting

Carrying Out	
– diligence, innovation, contribution to group (*)	30%
– peer assessment, poster	5%
Individual Report	50%
Group Oral Presentation	15%
Total	**100%**

(*) as demonstrated throughout the project from interaction with supervisors and during weekly meetings

5.3 Project Presentation

The project presentations for all groups will be held on **Thursday June 19th 2025**. Each group's technical presentation should be 1.5 hours long and will be followed by a 30-minute Q&A session.

The presentation should provide an overview of the students' proposed solution to the project brief. In particular, it should detail the students' approach to the problem, identify major design decisions and the reasoning behind them, present and discuss major results and assess the efficacy of the proposed solution. The most appropriate format, content and breakdown of the presentation are to be determined by the students.

Additionally, each group will have to prepare a 5-minute pitch of their concept. Departmental staff, other students and members of the Departments industrial advisory board will be in attendance. The presentation should present an overview of the work done by each GDP group but will not be assessed.

Students may use the Audio-Visual facilities in any of the lecture rooms at any time when free to practice your presentation (room timetables on doors).

5.4 Presentation Marking

Presentations are marked on the following criteria -

Quality of viewfoils: 20%
Clarity of presentation: 20%
Explanation of Problems: 20%
Technical Challenge & Solution: 20%
Demonstrated level of Group Work: 20%

5.5 Project Report Guidelines

Each student is required to present the results of their research project in a report, the deadline for which is **Monday June 16th 2025**.

The report should detail your contribution to the project and address and justify design choices made throughout the project. A typical report should include an abstract, brief introduction and theory, results, discussion and conclusion sections as necessary to best present your work. A standard template is not provided; instead a common template for the report should be chosen and used by the entire group.

The report should be no longer than 12 pages. The cover page, Tables of Contents and Appendices are not included in such limit. Markers are instructed to not read beyond the page limit and ignore excess pages when marking.

Advice on report writing can be found in the Student Handbook, but students are advised to consult projects from previous years to see the standard expected. In particular, look at reports which in the past have won prizes. Students can view past GDP reports on Sharepoint.

There are penalties for late submission of project work on exactly the same basis as those for coursework (see "Coursework Submission Penalty Scheme" section in your student handbook).

5.6 Carrying Out Marking

The student is also assessed on their conduct (Carrying Out) of the project. This mark is produced at the end of the project. A student who communicates well with the supervisor and group, attends meeting, takes initiative to think about problems before taking them to the supervisor, reads around the topic area, is careful and efficient with calculations, programming and/or experimental procedure, and (most importantly) takes an interest in and makes an important contribution to the project, can expect to get an very good Carrying Out mark.

The peer-assessment component will be based on peer-reviews to be completed on a weekly basis by each project participant

5.7 Mark Schemes

To ensure uniformity of assessment a standard system of grading is to be used. This is based on the criteria given in the mark schemes attached to the end of this document.

6. Project Submissions

Submission Information for Project Reports and Posters

Final Report – Monday 16th June 2025

- *ONE* pdf file of your individual report to be submitted via your report submission box on Blackboard.

Executive Summary – Monday 16th June 2025

- *ONE* pdf file of the group's design specification document to be submitted via the Executive Summary submission box on Blackboard (no hard copies to be submitted with your reports).

Group Poster – Thursday 19th June 2025

- *ONE* pdf of an A1-size poster to be submitted via the Poster submission box on Blackboard. You may wish to print a copy for your group presentation and/or pitch.

Carry-Out Mark-Scheme

Criteria		Fail 0-39%	Poor/Satisfactory 40-59%	Good 60-69%	Excellent 70-89%	Outstanding 90-100%
Contribution to group (20%)	Group working dynamic	Demonstrated little to no ability to work as part of a group	Demonstrated limited ability to work as part of a group	Demonstrated good ability to work as part of a group	Demonstrated excellent ability to work as part of a group	Very little could be done to better it in any way
	Participation in group discussions	Very limited participation in group discussion	Some participation in group discussion	Good participation in group discussion	Active participation in group discussion	
Diligence (30%)	Work standard	Work presented at the progress meetings is of a poor standard	Work presented at the progress meetings is of satisfactory standard	Work presented at the progress meetings is of good standard	Work presented at the progress meetings is of excellent standard	
	Effort shown in the project work	No significant effort was evident throughout the duration of the project	Some effort was evident throughout the duration of the project	Good effort was evident throughout the duration of the project	Excellent effort was evident throughout the duration of the project	
Reporting (20%)	Communication skills at weekly presentations	Poor communication skills (voice unclear, no eye contact, mostly reading from the presentation slide/note)	Limited communication skills (Sometimes clear voice but co nstantly reading from slides/notes)	Very good communication skills (Clear voice, but falls short on engaging with the audience)	Excellent communication skills evident (Clear voice, engaging the audience)	
	Weekly presentation slides/ visuals	Poor slides/visuals, too much text and sometimes not legible, little use of graphics/effects, no sources cited	Slides/visuals are legible but inconsistent, some use of graphics/effects, little or no sources cited	Slides/visuals are legible, good use of graphics/ effects, sometimes cited the sources	Excellent, attractive slides, text/figures are legible, graphics/effects used to enhance presentation, sources are cited well	
Technical Skills (15%)	Identify which computational, analytical, experimental tools to use/develop		Limited ability to identify the appropriate computational, analytical or experimental tools to use/develop	Some ability to identify the appropriate computational, analytical or experimental tools to use/develop	Excellent ability to identify the appropriate computational, analytical or experimental tools to use/develop	
	Support required for technical matters		Too much support required. No evidence of independent working	Some support required for technical matters (Students able to work independently with some support from Supervisor)	Minimal support required for technical matters (Students able to work independently without relying too much on Supervisor's input)	
Originality (10%)	Original ideas		Ideas and approaches are both traditional and predictable. No originality evident	Some innovative approaches were employed in the project	Creative and innovative ideas were employed in the project	
	Methods		Methods used are too simplistic for the project	Methods used were mostly appropriate for the project	Methods used were well reasoned and entirely appropriate for the project	

Report Mark-Scheme

Criteria		0-39% Fail	40-49% Poor	50-59% Satisfactory	60-69% Good	70-89% Excellent	90-100% Outstanding
Structure of report (15%)	Abstract quality	No abstract presented.	Some form of abstract presented.	Abstract present but of poor quality.	Good abstract with minor deficiencies.	Complete and very good abstract.	Very little could be done to better it in any way
	Include appropriate sections			Sections missing (within reasonable expectations of complete report given content)	All reasonable sections (relevant to content) present but not properly balanced.	All reasonable sections (relevant to content) present and balanced.	
	Introduction (*providing overall context and objectives of the report*)	No introduction and missing the aims and objectives	Poorly written introduction, not stating the aims and objectives.	Poorly written, not giving context/ description to the overall design concepts, and poorly stated/ missing aims and objectives	Introduction provided giving some context of the design for the report and some aims/objectives explicitly stated	Excellent introduction, setting a clear context of the overall design in the report, stating clear/achievable aims and objectives	
	Methodology/ approach	No explanation of methods/ approach used to carry out the project/task.	Limited/almost no explanation of methods/ approach used to carry out the project/task.	Very vague explanation of the methods/approach used to carry out the project/task.	Good explanation of the methods used to carry out the project/task but choice of method not justified.	Clear and concise explanation of the methods used in the project/task, with justification as to why they are the most appropriate for the task.	
Validity of results (25%)	Appropriate table/ graphs used to present data/ results	No actual data presented (relevant to content)	Data not presented in most appropriate format (relevant to content)	Data partly presented in most appropriate format (relevant to content) but serious deficiencies.	Data presented in a reasonable format with some deficiencies.	Data presented in most suitable format (tables/graphs/etc), in a clear and concise manner.	
	Relevancy of data/results	Data/results are irrelevant to project.	Limited data/results present with very limited relevance to project.	Some data/results present with limited relevance to objective of report.	Most data/results present relevant to the objective but with some deficiencies.	Data/results presented in perfect alignment with objective of report.	
	Validity of data/ results	Data/results are not justified by means of appropriate calculations. Results presented are incorrect.	Data/results are not justified by means of appropriate calculations. Results presented are not all correct.	Data/results are partly justified by means of appropriate calculations. Questionable validity of results presented.	Data/results are mostly justified by means of appropriate calculations. Results looked partly valid.	Data/results are justified by means of appropriate calculations. Results are mostly valid/sensible.	
Discussion of results (25%)	Discussion of results	Results presented but not discussed.	Results presented with very limited discussion.	Results presented and described but not analysed/justified.	Results presented and described but with some deficiencies in the arguments.	Results discussed/ justified correctly in the most suitable manner relevant to content/objectives of report.	
	Report back up with literature	Discussions/arguments not linked to literature findings.	Poorly use of literature to support discussions/arguments.	Insufficient use of literature to support discussions/arguments.	Some reference to the literature in discussion/arguments with deficiencies.	Discussions/argument s appropriately linked and supported by literature.	
	Clarity of arguments in discussion	Discussion arguments extremely confusing and/or hard to follow.	Discussion arguments not very clear/ need a lot of effort to understand/ follow.	Discussion arguments not very clear/ need some effort to understand/follow.	Discussion arguments mostly ok but some areas unclear.	Discussion arguments clear and well explained, easy to follow.	
Closing remarks/ summary (15%)	Limitations and challenges	No discussion of limitations/challenges of the project.	Very limited discussion of limitations/challenges of the project.	Some discussion of limitations/challenges of the project.	Good discussion of limitations/challenges of the project.	Excellent discussion of limitation/challenges of the project.	
	Future design iterations and improvements	No reference to how the results may be improved with future work.	Very limited reference to how the results may be improved with future work.	Some discussions to how the results may be improved with future work.	Good discussions to how the results may be improved with future work.	Excellent discussions to how the results may be improved with future work.	
References/ Bibliography (10%)	Format/ style of reference used	References and citations not in correct Vancouver style.	Vancouver style used but with numerous errors.	Vancouver style used but with many errors.	Vancouver style used but with some errors.	Correct Vancouver style/format used.	
	Number and variety of reference used/ quality of references	No references used in report.	Limited references and of poor quality.	Some references, but of poor quality.	Good list of references but not from varied range of quality sources.	An excellent range of references, correctly supporting the text and used appropriately, from a varied range of quality sources.	
Formatting/ presentation of report (10%)	Spelling/ grammar	Numerous spelling/grammatical errors indicating no effort.	Many spelling/grammatical errors.	Some spelling/grammatical errors.	Very minor/grammatical errors.	No spelling/grammatical error.	
	Format of report	Very poorly formatted report indicating no effort.	Poorly formatted report indicating limited effort.	Many mistakes in the formatting of the report.	Some mistakes in the formatting but mostly following style guide.	Report correctly formatted throughout.	
	Format of Tables/ Figures/ Equations	No captions for the figures, tables, and equations.	Significant issues with captions and presentations of figures, tables, and equations.	Some issues with captions and presentations of figures, tables, and equations.	Very minor issues with captions and presentations of figures, tables, and equations.	All figures, tables, and equations correctly labelled and referred.	

Presentation Mark-Scheme

Criteria	**0-39% Fail**	**40-49% Poor**	**50-59% Satisfactory**	**60-69% Good**	**70-89% Excellent**	**90-100% Outstanding**
Explanation of Problems [20%]	A minimal explanation of the task and technical problems that were addressed, both at system and subsystem level.	Some explanation of the task and technical problems that were addressed, lacking in detail and overlooking with the system and subsystem level implications.	Most key problems have been addressed. Issues with the focus being wrongly skewed towards the system or subsystem level. The context, relative importance and impact are identified and discussed but with significant gaps or issues.	Problems that had to be addressed are mostly presented and explained. Minor issues with the focus being wrongly skewed towards the system or subsystem level. Minor issues with their context, relative importance and impact being identified and discussed.	Problems that had to be addressed at system and subsystem level are effectively presented and explained. Their context, relative importance and impact is correctly identified and discussed.	Very little could be done to better it in any way
Technical Challenge & Solutions [20%]	A minimally challenging technical task with no significant discussion provided for the choice of methods. Results presented are incomplete and/or highly suspect.	A moderately challenging technical task to be completed. The rationale for design decisions and methods is provided but with significant shortcomings. Results presented are not entirely relevant, complete or correct.	A moderately challenging technical task to be completed. The rationale for design decisions and methods is provided but with several shortcomings. Results presented are not entirely relevant, complete or correct.	A challenging technical task to be completed. The rationale for design decisions and methods is provided but with minor shortcomings. Results presented are relevant, mostly complete and correct.	A highly challenging technical task to be completed. A clear and correct rationale is provided for the design decision and methods discussed. Results presented are relevant, complete and correct. An understanding of uncertainty and/or limitations and their impact on the broader design is clearly evident.	
Demonstrated Level of Groupwork [20%]	A disjointed presentation, demonstrating no significant interaction between individuals in the completion of the work presented.	A somewhat disjointed presentation, demonstrating some interaction between individuals in the completion of the work presented. How the concerns of different technical specialists were balanced is barely touched upon.	A presentation that in places demonstrates the impact of group member collaboration. Falls well short of demonstrating how some key stakeholder requirements were considered or appropriately balanced.	A mostly coherent presentation that demonstrates how group members communicated and collaborated. Falls short of demonstrating how all stakeholder requirements were considered or appropriately balanced.	A coherent presentation that clearly demonstrates how group members effectively communicated and collaborated to reach a globally optimal solution that balances all stakeholder requirements and concerns.	
Clarity of Presentation [20%]	Poor presentation skills. Clumsy, disjointed, difficult to follow, or dull. Significantly under or over time.	Not always clear or easy to follow, unimaginative and unengaging. Significantly over time. Material fairly disorganised and rushed.	Effectively conveys intended ideas, but is sometimes unclear or clumsy. More or less right length, but some material not covered properly as a result or relative focus on sections not well balanced.	A clear, lively and engaging presentation. Good engagement with the audience. Well organised, but some issues with the balance of focus on sections. More or less to time.	An excellent presentation. Clear, lively and engaging, presenting complex information in a logical narrative order. The focus on different aspects of the design was highly appropriate for their relative importance in the overall design. Perfectly timed.	
Quality of Viewfoils [20%]	Poor or no use of visual media to support the presentation. Content exceptionally difficult to understand.	Visual media used to support the presentation, however there is significant room for improvement. Lack of appropriate detail with significant issues relating to layout, fonts and colouring employed	Good use of visual media to support the presentation, several improvements needed. Content presented to appropriate detail with appropriate layout, fonts and colouring employed, with significant issues	Very good use of visual media to support the presentation, minor improvements needed. Most content presented to appropriate detail with appropriate layout, fonts and colouring employed, with some minor issues.	Excellent use of visual media to support the presentation. All content presented to appropriate detail with appropriate layout, fonts and colouring employed.	